

AWI-ESM3 ↔ PISM moving-cavity coupling

the investigation log

Single living document. Superseded claims are struck through, not deleted.

Last updated: 2026-07-13

Abstract

ROOT CAUSE FOUND (2026-07-14): the salt plume parameterisation was OFF (`spp = .false.`). `spp` distributes the brine rejected when sea ice grows *downward* through the water column instead of leaving it at the surface. That is how High Salinity Shelf Water forms. Without it no dense shelf water forms, warm deep water floods the Antarctic shelf and then the ice-shelf cavities. Our cavity sat **≈ 1.2 K above freezing**, melted the shelves at **≈ 28 m/yr** ($30\times$ **observed**), and PISM duly collapsed — shedding **24% of its floating area per decade**. **The catastrophic ice geometry the ocean kept failing on was manufactured by our own melt**. Xiaojie had already demonstrated this in standalone CORE3 (§4).

Three root causes we published and later falsified are struck through below. The moving-cavity blowup itself is real but secondary: the cavity *geometry* is fine (a cold start on it is stable at the production timestep); the killer is the *seam* in our remapped restart, where patched values at the changed nodes meet a fully evolved dynamical state everywhere else.

Contents

1	Blocker I — OIFS step-0 coastline crash (RESOLVED)	1
2	Blocker II — FESOM cavity-margin blowup	1
2.1	Two root causes we got wrong	1
2.2	Every hypothesis, and the experiment that killed it	2
2.3	The decisive control: a cold start on PISM's cavity is stable	2
2.4	What is actually wrong: the dynamics seam	2
3	The real increment test (2026-07-13)	3
3.1	The finding: PISM is collapsing	3
3.2	This is not good news	4
4	ROOT CAUSE: the salt plume parameterisation was off	4
4.1	The chain	4
4.2	The evidence	5
4.3	The fix	5
5	Where to go from here	5
6	Bugs found and fixed along the way	6
6.1	In the remap	6
6.2	In the coupling plumbing	6

1 Blocker I — OIFS step-0 coastline crash (RESOLVED)

A floating-point NaN struck exactly the coastline cells whose land-sea type flipped. `ocp-tool` half-converted a flipped cell (only mask + soil type), OIFS re-ingested the inconsistency from the re-generated ICMGG and NaN'd in moist physics. Fixed by a masked nearest-neighbour rebuild of each flipped cell's full column, plus the ICMGG re-ingest (PUTALLFLDS_M in `reresf_part2.F90`). A recurring `voskin` floating-overflow variant shows up state-dependently and was traced to the native-pipeline ICMGG coastal lsm.

2 Blocker II — FESOM cavity-margin blowup

2.1 Two root causes we got wrong

FALSIFIED #1 — “a geostrophically-unbalanced remapped state”. ~~The mesh change plants fresh cavity fronts against salty open water but zeroes the balancing current; initialise the changed-cell velocity in thermal-wind balance instead of zero. That fix was implemented (per-level $\mathbf{u}_g = (1/f\rho_0) \hat{z} \times \nabla p$, using FESOM's own cavity pressure convention) and failed its own offline gate: the discrete ∇p at grid-scale fronts is *noisier* than the nearest-neighbour fill. It ships opt-in only (REMAP_GEOSTROPHIC=1), off by default.~~

FALSIFIED #2 — “the sub-ice free-surface BC violation”. ~~FESOM keeps $\eta \equiv 0$ under ice, yet the remap gave the 987 newly-sub-ice nodes their old open-ocean ssh (≈ -1.6 m); fixing that will cure the blowup. A real bug, and fixed (esm_tools b17fb5c8) — but not the cure: the fixed run still died on day 4.4.~~

2.2 Every hypothesis, and the experiment that killed it

Hypothesis	Experiment	Verdict
Atmosphere / OASIS / coupling	ocean-only standalone FESOM reproduces the crash exactly (day 4, CFLz 9.24)	ruled out
CORE2 forcing shock (standalone artefact)	full-mesh control, <i>same</i> forcing + native restart: stable 21 d, worst CFLz 2.61	ruled out
Remap corrupting the state	remapped vs. source over the 204970 unchanged nodes: $\max \Delta = \mathbf{0.000}$ for temp, salt, ssh, hnode	ruled out (bit-exact)
The restart <i>fill values</i> at changed nodes	five independent fill strategies (donor, coherent-column, shelf-base hold, NN velocity, full balance hygiene)	ruled out (all die $\approx$ day 4)
Pathological thin water columns	submesh min wet column 30 m vs. mother's 25 m; zero 1–2-layer columns	ruled out (sub-mesh is <i>cleaner</i>)
Basal melt flux shock	<code>cavity_flux_ramp_days</code> implemented; melt effectively OFF still dies (mstep 328)	ruled out
Timestep / unresolved w	1200 s: CFLz kill day 4.0. 60 s: 0 CFLz warnings, dies day 4.4 via η_n . 120 s: dies day 8.9	ruled out

Hypothesis	Experiment	Verdict
Sub-ice free-surface BC Cavity geometry itself	real bug, fixed; run still dies day 4.4 cold start on the same PISM sub-mesh @1200 s: STABLE, day 13+, worst CFLz 3.44	not the cure ruled out

2.3 The decisive control: a cold start on PISM’s cavity is stable

Same submesh, same forcing, same production timestep — only the initial state differs:

Run	Δt	Initial state	Outcome
full-mesh control	1200 s	native spun-up CORE3 restart	stable 21 d, worst CFLz 2.61
PISM cavity, cold	1200 s	cold (climatology, at rest)	stable 13+ d, worst CFLz 3.44
PISM cavity, warm	1200 s	our remapped restart	dies day 4.0 , CFLz 9.24
PISM cavity, warm	120 s	our remapped restart	dies day 8.9 (η_n)
PISM cavity, warm	60 s	our remapped restart	dies day 4.4 (η_n , 0 CFLz)

The ocean has no difficulty living in PISM’s cavity at the production timestep. Whatever is wrong is in *our restart*, not in the mesh, the physics, or the coupling.

2.4 What is actually wrong: the dynamics seam

The remap produces an internally inconsistent state: on the **unchanged nodes (98%)** it reproduces the evolved restart **bit-exactly** (a fully developed, balanced dynamic state: $\mathbf{u}$, η , AB history); on the **changed nodes** it substitutes *patched* values. The two regions are mutually incompatible, and the **seam between them** is the instability. Zeroing the dynamics globally removes the crash (the ocean re-derives its own balanced flow from rest) — which is exactly why all five fill strategies failed: each varied the *patch* while leaving the *seam*. Zero-dynamics is not a production fix: discarding the ocean’s circulation every leg is unacceptable.

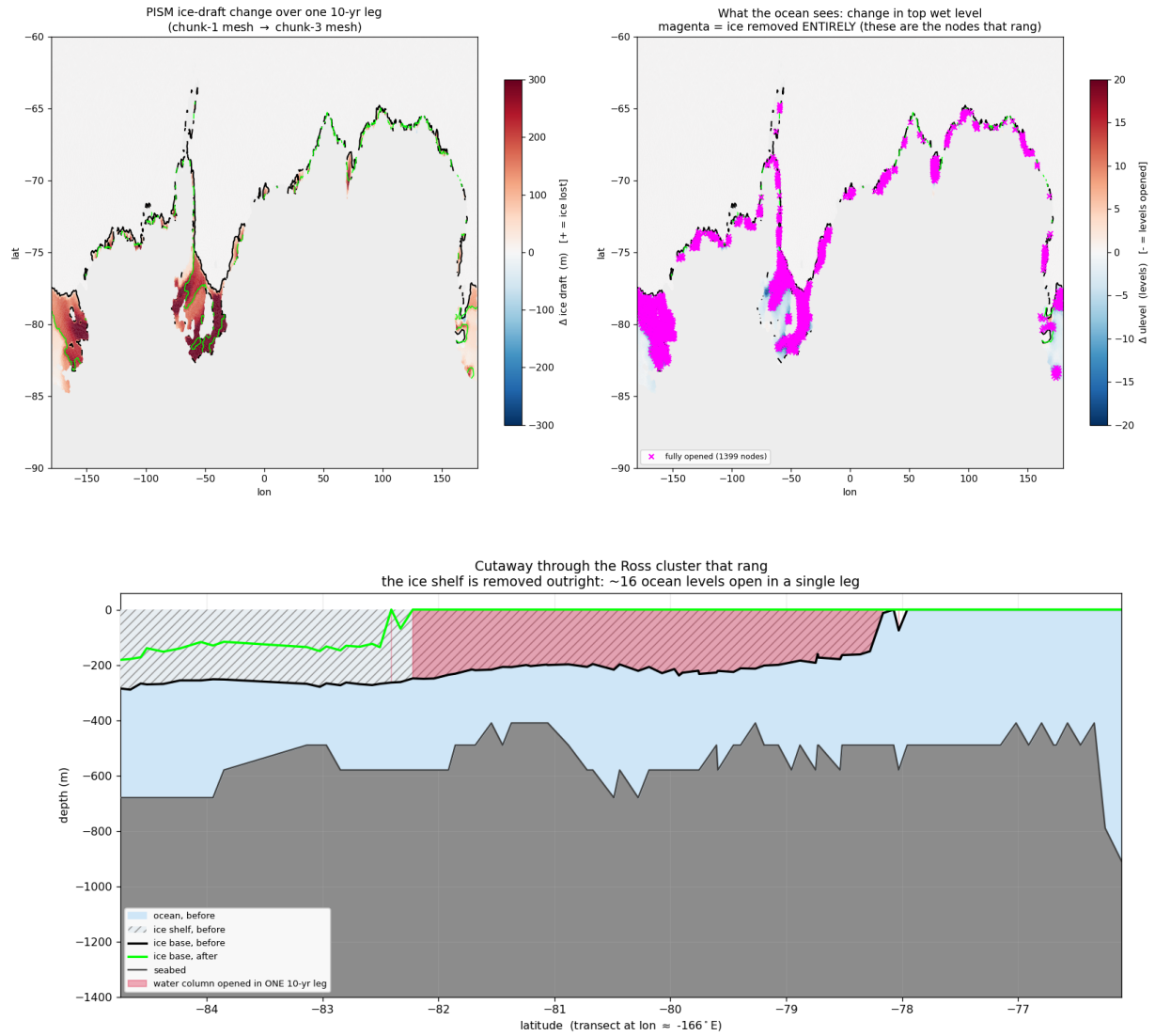
3 The real increment test (2026-07-13)

Cold-starting the ocean on PISM’s cavity sidesteps the one-off CORE3(observed)→PISM swap ($\max|\Delta\text{draft}| = 2711$ m) entirely. That was done: chunk 1 ran a clean model year, PISM ran its 10 years, and the remap was finally asked to carry *a year of spun-up ocean dynamics across a real PISM increment*. It crashed on day 15.7.

	one-off swap (CORE3→PISM)	real increment
CFLz over time	4.46 → 6.43 → 7.25 → 9.24 monotonic runaway	5.26 → 5.78 → 5.40 → 4.15 peaks, then decays
died at	day ~4	day 15.7
η_n NaN	yes	none — all 209 172 nodes finite
$ \eta > 3$ nodes	3611, basin-wide	17 , one cluster, all changed nodes

All 17 ringing nodes share one signature: $\text{ulev } 17 \rightarrow 1$ — **16 ocean levels opening at once**, i.e. the ice shelf above them removed *entirely*. Their η alternates in sign (+24.4, −17.9, +15.9, +8.3): a grid-scale **checkerboard**, not a basin mode.

3.1 The finding: PISM is collapsing



The cutaway is the clearest statement of the problem: the ice base goes from $-200 \dots -280$ m to **zero (open ocean)** across the whole span -82.2° to -78.2° latitude. **The Ross ice-shelf front retreats ≈ 440 km in ONE 10-yr leg.** The plan view shows this is not a Ross anomaly: the 1399 fully-opened nodes (magenta) ring the *entire* Antarctic margin.

changed nodes this leg	3248	(1.55% of mesh)
$ \Delta \text{ulev} \geq 10$	1582	
nodes with the ice removed ENTIRELY	1399	
mean $ \Delta \text{draft} $	2.9 m	← meaningless
max $ \Delta \text{draft} $	923 m	
PISM floating cells	24 815 $\rightarrow$ 20 093	(-24% in 10 yr)

FALSIFIED #3 — “the per-leg increment is gentle”. Production never has to survive the one-off swap; the per-leg PISM increment is two orders of magnitude smaller (mean $|\Delta \text{draft}| \approx 22$ m) and therefore absorbable. **Wrong, and wrong in a way that mattered** — it motivated the entire “sidestep the swap and the increments will be fine” strategy. The mean is diluted by the 98% of the domain that does not change; the distribution is wildly skewed. PISM is *not* in equilibrium.

3.2 This is not good news

It is tempting to read the numbers optimistically: only 17 of 1399 fully-opened nodes rang; the ocean absorbed $\sim 99\%$; the CFLz transient peaked and *decayed*. All true, and it does mean the remap machinery is performing better than a “crashed at day 15” headline suggests.

But the honest reading is negative. The run still crashed. What we have learned is that **our ocean reacts violently to ice-shelf and cavity changes mid-run**: a handful of columns where the ice vanishes outright is enough to kill a leg. A coupled setup in which the ice sheet is *allowed* to change its cavity every 10 years cannot be one leg’s worth of extreme nodes away from failure. **Absorbing 99% of the change is not a passing grade when the remaining 1% is fatal.**

4 ROOT CAUSE: the salt plume parameterisation was off

4.1 The chain

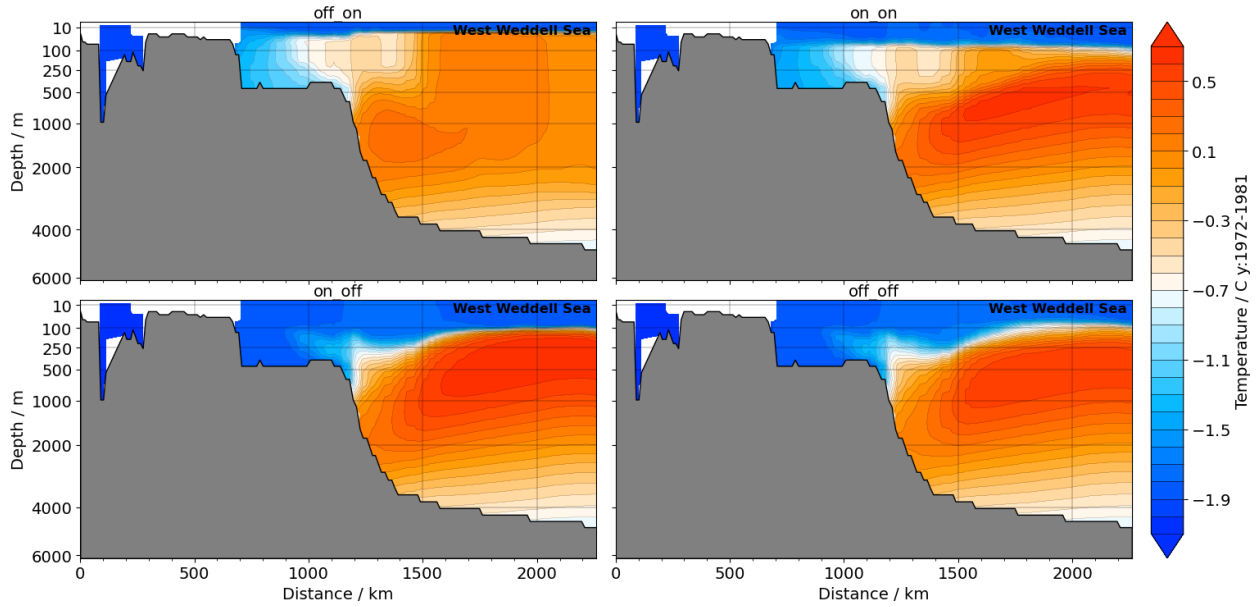
1. `spp = .false.` $\Rightarrow$ brine rejected by growing sea ice stays in the surface layer and is mixed away.
2. No brine plume $\Rightarrow$ **no High Salinity Shelf Water forms.**
3. No HSSW $\Rightarrow$ nothing excludes the warm deep water, which floods the Antarctic shelf and then the cavities.
4. Cavity sits $\approx 1.2\text{ K}$ **above freezing** instead of $\approx 0.2\text{ K}$.
5. FESOM melts the shelves at $\approx 28\text{ m/yr}$ (observed area-mean $\approx 0.8\text{ m/yr}$).
6. PISM, fed that, sheds **24% of its floating area per decade**: 1399 nodes lose their ice *entirely* in one leg, the Ross front retreats $\approx 440\text{ km}$.
7. The ocean is then asked to swallow that geometry change every leg — and fails.

The “PISM is collapsing” problem and the “the ocean cannot absorb the increment” problem are the same problem, and we caused both.

4.2 The evidence

The initial condition is irrelevant — the model heats the cavity itself. We prepared a chunk-1 restart with a properly cold cavity (thermal driving $+0.06\text{ K}$, from the spun-up CORE3 tracers). After *one model year* it had warmed to $+1.25\text{ K}$ — and the cold-started run converged to the same $+1.21\text{ K}$. The melt was identical (28.28 vs 28.29 m/yr). Initialising the cavity correctly buys nothing if the model then warms it back up.

Xiaojie had already shown this in standalone CORE3 (West Weddell transect, 1972–1981; `/home/a/a270234/my_work/20260407/temp_fesom_core3_WWS.ipynb`):



Top row (**spp ON**): the shelf is cold to the bottom and the warm deep water is pushed offshore and down. Bottom row (**spp OFF**, **our configuration**): the warm water sits right against the shelf break, ready to intrude. **spp** is the dominant switch.

4.3 The fix

spp: `".true."` in `namelist.oce &oce_dyn` (runscript `awiesm3_dyn_ocean_core3.yaml`). Note `use_momix` is already on by FESOM's default.

5 Where to go from here

1. **Re-run with `spp = .true.`** and confirm the chain unwinds: cavity thermal driving should fall to $\approx 0.2\text{K}$, melt to order 1m/yr , PISM should stop collapsing, and the per-leg geometry increment should become genuinely small — which is what the remap was always designed for.
2. **Then re-test the increment.** With a sane increment, the dynamics seam (§2.4) may never be exercised hard enough to matter. If it still bites, the remaining case is narrow and well defined: a column where the ice vanishes entirely, handed donor water, zero velocity, and a free surface that cannot match its still-cavity neighbours.
3. **Independently, fix the PISM ocean forcing.** `coupling_ocean2pism.functions` averages FESOM levels `1–20 = 0–410 m`, **starting at the surface** (`-sellevidx,1/20`) and hands that to PISM's `-ocean th`, a *local boundary-layer* scheme that needs the water *in contact with the ice base*. The correct code (`-sellevel,150/600`) sits right there, disabled behind if `[ 0 -eq 1 ]`. This is a second, independent error — it did not cause the collapse (FESOM's own melt was equally catastrophic) but it is wrong and should go.
4. **Consider passing FESOM's melt directly** (`-ocean given`). We run an explicit cavity and FESOM already computes basal melt; having PISM recompute it with its own three-equation scheme means two implementations disagreeing about the same interface, and ice and ocean seeing different melt.

6 Bugs found and fixed along the way

6.1 In the remap

Unbounded extrapolation below the old seabed (esm_tools eef4f6db). The first attempt at the increment leg died at **mstep 1**: the remapped restart contained -45.02°C , a value present *nowhere* in the source (wet range $[-2.32, 41.72]$, dry cells 0.0). The remap *manufactured* it. The column shows the mechanism: 1.22, -0.99 , -3.19 , -5.39 , -8.69 , -13.09 , -18.6 , -25.2 , -34.01 , **-45.02** — smooth and accelerating. For columns that get *deeper*, the remap fitted a straight line through the last two old levels and continued it downward with no clamp. Fixed by taking such water from the nearest old node actually wet at that depth. **Never hit before** because every previous remap went mother $\rightarrow$ submesh (a strict *subset*); ice *retreating* grows the mesh and exercises this path for the first time.

Also: sub-ice ssh/hbar zeroing (b17fb5c8); a **size guard** so the remap refuses a restart that does not match the old mesh, instead of silently writing garbage (3b44a704).

6.2 In the coupling plumbing

Making chunk 1 a *submesh* leg pushed a submesh leg through `couple_out` for the first time. Every previous run either ran chunk 1 on the *full mesh* (where these paths are trivially correct) or died early in chunk 3 (before reaching the end of a leg). That exposed:

- **OASIS GSSPOS abort at ~day 300**. Its conservation correction is built from the summed source/target ratio; when the summed target residual is ≈ 0 while the source residual is small-but-nonzero, it explodes. Only the heat-to-ice flux (`A_Q_ice  $\rightarrow$  heat_ico`) used it. Fixed: new `gauswgt_gsmart` transform (3e94d4da).
- **The namcouple feom dimension had NEVER been patched, on any leg**. `fix_namcouple_feom_dim` read `MAXMESH_DIR_fesom`, which is assigned *inside* `build_submesh` — a different subjob's shell. Always unset $\Rightarrow$ guard refused $\Rightarrow$ silent no-op. Fixed (3cf29e9e).
- **fesom2ice loaded the static full mesh** while FESOM's output is on the submesh (broadcast error) (63cfcc91).
- **previous_submesh is read but never created** — anywhere. Every leg from chunk 5 on would have remapped against the wrong mesh (3b44a704).
- **A failed couple_in did not stop the workflow**. The coupling function is backgrounded, the observe job is not its parent (so it can never see the exit code), and `observe.py` returns an *empty* error list for every non-compute job. The next leg started on an un-remapped restart. Fixed with a `.coupling_failed` sentinel (3b44a704).

The pattern: the guards were right, and the plumbing around them defeated them. `remap_oasis_restart` correctly refused a mis-sized restart — and the run proceeded anyway.

7 Assets

- **A 2-minute offline reproducer:** standalone ocean-only FESOM on the PISM-cavity sub-mesh, 14 nodes, crashes in ~ 2 min (vs. the 45-node coupled system). Requires the `FESOM_COUPLED=OFF` build at the current SHA.
- **A stable control:** full mesh + native restart + same forcing (21 d, CFLz 2.61).

- **Chunk-1 artifacts pooled** at `input/fesom2/pism_cavity_ini/` (540 MB), so chunk 1 needs no `couple_in` and `esm_runscripts` can be run straight from the login node.
- `cavity_flux_ramp_days` (FESOM `&run_config`, default 0 = off, bit-identical).
- `verify_balance.py`: offline balance/stability gate ($\min N^2$, $\max |\nabla \cdot (H\bar{\mathbf{u}})|$, CFL, Ri) — it caught the geostrophic ∇p init being worse than the NN baseline *before* it cost a model run.

8 Method note

Three root causes were asserted with more confidence than the evidence supported, and all three were later falsified by a control that was cheap and should have come *first*:

1. “geostrophic imbalance” — killed by the **cold start** (is it the geometry or our state?);
2. “the sub-ice ssh BC” — killed by simply **running the fix**;
3. “the increment is gentle (≈ 22 m)” — killed by **plotting the distribution** instead of trusting a mean over a domain that is 98% unchanged.

The third is the sharpest: the 22 m figure was computed correctly and reported honestly, and it was still misleading enough to steer the strategy for a day. **Two figures did in ten minutes what a summary statistic obscured for a day.**